AAKASH SIRICILLA

aakash.siricilla02@gmail.com ◇ +1 (551)-375-6945 ◇ LinkedIn ◇ Portfolio ◇ Github

EDUCATION

New Jersey Institute Of Technology, Master of Science in Computer Science — Sep 2023 – May 2025
Coursework: Distributed Systems, Deep Learning, Machine Learning & MLOps — **GPA: 3.93/4.0**

Kommuri Pratap Reddy Institute Of Technology, B.Tech. Computer Science & Engineering — Aug 2019 – May 2023
Coursework: Cloud Computing, Data Structures & Algorithms, Operating Systems, Database Systems — **GPA: 8.55/10**

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, SQL (PostgreSQL, MySQL), C/C++
Frontend	React, Next.js, Tailwind CSS, HTML, CSS
Backend & APIs	Spring Boot, FastAPI, REST APIs, WebSockets, JWT Authentication, Async Processing
Databases & Messaging	PostgreSQL, MySQL, Redis, Qdrant, FAISS
Cloud & DevOps	Docker, Kubernetes, CI/CD, AWS, Git, GitHub Actions
Systems & Tools	Celery, MLflow, Prometheus, Grafana, Pandas, NumPy

WORK EXPERIENCE

New Jersey Institute of Technology — Mar 2025 – Present
Research Assistant — *Newark, NJ, USA*

- Optimized PostgreSQL and MySQL queries across 120K+ records by redesigning joins, refining retrieval logic, and improving indexing strategies, reducing query execution time by 40% and improving backend responsiveness for internal research applications.
- Automated Python-based ETL and preprocessing pipelines for recurring datasets, reducing manual processing effort by 60% and improving the consistency and reliability of downstream reporting workflows.
- Built internal data pipelines and reporting dashboards to track operational and model performance metrics, shortening reporting turnaround time by 30% and improving visibility into system performance for research stakeholders.

Cognier Insights — Apr 2023 – Sep 2023
Software Engineer — *Hyderabad, India*

- Built internal CRM and reporting features using Java, Spring Boot, React, and PostgreSQL, including customer profile management, dashboard views, and workflow automation tools, reducing manual operations by 35%.
- Designed and developed REST APIs, service-layer validation, and PostgreSQL-backed data access logic for data-intensive workflows, improving retrieval performance by 25% and increasing reliability of internal transaction flows.
- Created reusable React UI components and integrated them with Spring Boot services, reducing feature delivery time by 30% and improving consistency and maintainability across shared frontend modules.

PROJECTS

SyncStream - Real-Time Collaborative Streaming Platform | Java, Spring Boot, Next.js, WebSockets, PostgreSQL, Docker | GitHub

- Built a full-stack real-time collaboration platform with invite-based session flows, persistent room state, and low-latency synchronization between users using Spring Boot, WebSockets, and Next.js.
- Dockerized frontend and backend services, structured the project for local and cloud deployment, and implemented scalable client-server communication patterns for live session updates.

DocuMindAI - Enterprise Document Intelligence Platform | FastAPI, React, Qdrant, Redis, Docker, MLflow | GitHub

- Developed a scalable full-stack document platform with asynchronous ingestion, semantic search, and contextual chat, reducing document lookup time by nearly 70%.
- Implemented hybrid retrieval, background task execution with Celery and Redis, and modular service orchestration to support multi-document workloads and extensible backend architecture.

SentinelAI - Real-Time Network Monitoring Platform | Python, FastAPI, Redis, TimescaleDB, Next.js, Docker | GitHub

- Engineered a real-time event processing pipeline with streaming analytics, Redis-backed state management, and sub-500ms request handling for high-volume anomaly detection workflows.
- Built a responsive monitoring dashboard with WebSocket-powered updates, replay controls, and live analytics views, improving visibility and investigation speed for operational events.